

Understanding the Effects of Spatialized Audio Cues On Spatial Knowledge Construction in Virtual Reality

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ABSTRACT

Accuracy of spatial knowledge in virtual reality (VR) becomes increasingly crucial as more VR applications directly involve its use. Yet, the vergence-accommodation conflict (VAC) in head-mounted displays (HMDs) causes inaccurate spatial perception in VR. Some existing methods of addressing the depth perception issue in VR are computationally intense, require custom hardware, or subject to limitations in effectiveness. We propose integration of spatial audio in VR as an intuitive feature with lower financial and computational cost to better support spatial perception, which leads to spatial knowledge construction. This study explored the impact of spatial audio cues in VR on spatial mapping acuity in the context of an enclosed virtual environment. Results indicated spatial audio cues are naturally incorporated by humans to significantly improve human spatial knowledge accuracy, demonstrating for VR a suitable method of facilitating more spatially precise knowledge of an entire space.

Author Keywords

Spatial knowledge; virtual reality; spatial audio; head-mounted display.

ACM Classification Keywords

H.5.m. Information interfaces and presentation (e.g., HCI): Miscellaneous;

INTRODUCTION

As more uses for VR directly involving accurate spatial knowledge are developed, its ability to foster the acuity of spatial perception becomes increasingly important. VR is currently used as an interface for forms of spatial knowledge training such as for dentistry and the reduction of cognitive decline [6, 15]. It is also used for servicing remote evaluation of interior spaces (e.g. homes or office

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spaces) where the physical space's spatial qualities can deeply affect human emotion and behavior [18, 19]. Hence, even subtle differences in the precision of spatial mapping can strongly effect a user's experience of the environment.

A user's more precise spatial knowledge of the environment also translates to the opportunity to perform more complex operations in VR, as humans use that knowledge to inform their planning and interactions with the environment. Consequently, to improve spatial knowledge in VR means to broaden the scope of possible activities supported by it.

The sensory resources used by humans to acquire spatial knowledge must be delivered as naturally and accurately as possible in order to facilitate spatial knowledge acquisition more closely to how it occurs in reality. Depth perception is a core element of human spatial perception, which fundamentally contributes to their spatial knowledge. However, vision, the primary sensory depth cue in VR, is inherently inaccurate due to the VAC problem of HMDs.

The vergence-accommodation conflict

The VAC problem decreases fusion accuracy of binocular imagery due to conflicting cues caused by compressing a 3D environment into a stereoscopic display. Human eyes attempt to focus on a point of interest that appears to be some distance away in the illusion of a 3D space, but are in reality focusing on a point on the HMD lens in front of them, resulting in conflicting neurological cues [20]. This decreases the accuracy of egocentric depth perception in VR [1], in turn affecting spatial knowledge accuracy. The saliency of spatial elements as reference cues critically affects the human process of constructing spatial knowledge, since these cues are used as reference nodes to anchor the self in cognitive space [7]. But due to VAC, visual cues provided in VR through HMDs are not reliable as accurate cues for spatial knowledge construction.

Methods of improving spatial perception

There are currently a few ways to mitigate the depth perception issue in VR. The most direct way of addressing this issue is through multi-focal light field displays, which involves creating a sampled light field to closely emulate how light from different sources is usually received by human retinas [20]. However, custom ocular gear (e.g. programmable liquid lenses) is necessary, scene rendering

is computationally intensive, and the power required heats up the head-mount lenses considerably [21]. Incongruent representation of visual and auditory spatial information in VR to compensate for depth perception underestimation [10] has been a less costly software-based implementation to improve depth perception, which was successful at longer ranges but worsened estimates at closer range. The study did not investigate effects on spatial representation but on isolated distance estimates, and expressed uncertainty about the effect of incongruence on spatial cognition in a virtual environment where multiple elements appear at various positions around the user [2].

Haptic feedback in VR has also been explored for the additional modality it adds to interaction. It is shown to provide “proximal” information (tactile stimuli are the result of direct contact in close proximity to the object of interest) useful for scaling-related aspects of spatial knowledge [17], but does not provide “distal” information cues which are more related to the overall spatial map and its orientation [9]. However, auditory cues do provide distal information due to their directional and intensity properties.

Audio as a Compensatory Cue

Past research conducted on the effect of multi-sensory input (tactile, olfactory, and audio) in addition to visual cues on object memory in a virtual environment (VE) found that the inclusion of all three non-visual cues improved memory [4]. Amongst the three, spatial audio has the lowest computation cost [8] and intuitively encodes spatial information [17]. Audio events are also processed by humans as salient cues. Research conducted on the role of spatial audio on VR has explored topics such as presence [3, 16] and accessibility for visually-impaired individuals in nonvisual virtual environments [11, 17], but presently none has clearly shown its effect on spatial mapping accuracy for general individuals specifically in audiovisual VR.

Compared to current methods of specifically compensating for distance perception error or using haptic modality to bolster proximal spatial knowledge, using spatial audio alone, which is already widely supported in VR development, to facilitate spatial cues could sufficiently augment human spatial mapping performance and understanding of a virtual environment.

In the following study, we investigated whether spatialized audio effectively enhances VR spatial knowledge learning by having subjects engage in self-directed exploration of virtual rooms with elements bearing spatialized or non-spatialized audio cues, then perform sketch mapping tasks to measure spatial knowledge recall.

METHOD

A within-subjects experiment was conducted on 26 participants (13 males, 13 females) from ages 20-27 with two conditions (spatialized and non-spatialized audio) counterbalanced by task order and VE arrangement. An indoor room was used as VE and exploration process was

left to participants to more closely simulate VR exploration in practice and promote results generalizability.

Experiment Material

Conditions were split between two Unity scenes with various pieces of furniture. Each had the same selection of furniture but different arrangement and wall color. Among the 12 pieces of furniture, 7 were unique pieces and 5 were of the same chair. This was done to simplify the spatial mapping task, which was to sketch the locations of the five chairs in each room. Sketch maps were chosen to measure spatial knowledge recall due to its past validation as a measure for cognitive mapping of a virtual environment [14]. A short audio recording with descriptive content was attached to each piece of furniture and audio source spatiality was manipulated via the Steam Audio SDK (<https://valvesoftware.github.io/steam-audio/>). The spatial audio condition was implemented through the SDK. The 3D ability of sound to vary in source direction and intensity based on user orientation and VE geometry were achieved via binaural rendering over head-related transfer functions and simulated physics-based sound propagation. Binaural audio orients source origin based on varying sound intensity between the two ears to provide directional cues, and physics simulation varies intensity of sound based on its reflection off surfaces in the VE, as well as whether its trajectory to the user is occluded by any objects in the scene. The non-spatial audio condition rendered sound with no sound localization information – regardless of audio source or user position/orientation, all sound was rendered at a constant volume monaurally in both ears. Hardware used were the HTC Vive HMD and controllers.

Ten questions selected from the Purdue Spatial Visualization Test (PSVT) for mental rotation [5], scaled for difficulty distribution and time allotment, were provided on paper to evaluate each participant’s spatial ability due to known individual differences in this dimension [12].

Procedure

Before engaging in VR, participants first completed the PSVT sample of ten questions on paper with a time limit of eight minutes. Then to help participants acclimate to VR before performing the tasks, they explored for five minutes the Google Earth VR demo. Each subject was then instructed to explore the first room (with one of the audio conditions) by playing each audio source at least once and to take as much time needed to familiarize themselves with the space. Each piece of furniture changed color after its audio was played for the first time to help participants keep track of their progress. They were told prior to exploration that they would be asked to map the locations of the five chairs in the room afterwards. After participants indicated they finished exploring, they were asked to sketch the chair locations and footprint to scale on a provided map of that room, which included the location and footprint of all non-chair furniture. The same process was then repeated for the second room with the remaining audio condition.

Measures

Scores (out of 10) were obtained from the PSVT sample for each participant were obtained to compare spatial ability. Spatial mapping task completion time and distance errors between the ground truth location and subject-drawn location of each chair were used as performance measures.

RESULTS

No mapping time difference between the two conditions ($F < 1$, n.s.). However, a statistically significant difference on mean distance error in the predicted direction was observed between the spatialized ($M = 0.69$, $SD = 0.31$) and non-spatialized ($M = 0.81$, $SD = 0.43$) audio conditions, $F[1,25] = 4.22$, $p \leq .05$. Distance error means, used here to measure accuracy of spatial mapping, were obtained by averaging error measurements for each of the five chairs.

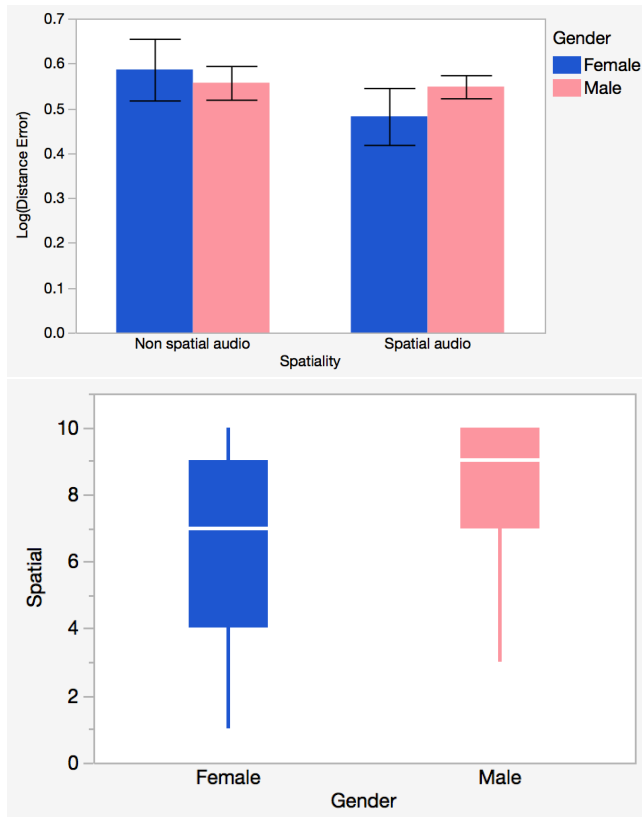


Figure 1. (Top) Normalized distance error by audio condition, gender compared side by side. Effect size (Cohen's d) was much larger for females ($d = 0.53$, medium) than for males ($d = 0.036$). (Bottom) Spatial ability score distributions by gender. Female distribution included more low-spatial ability participants than suggested by just the mean for each gender, indicated by the downwards skew of its box edge and tail.

Difference of distance error improvement by spatial audio (defined $de_{\Delta} = M_{spatial} - M_{non-spatial}$) between genders showed an evident trend of females ($M_{de_{\Delta}} = 0.20$, $SD = 0.28$) exhibiting more improved spatial mapping recall via spatial audio than for males ($M_{de_{\Delta}} = 0.023$, $SD = 0.26$), $F[1,25] = 2.64$, $p = 0.12$. Figure 1 displays measured gender differences in the collected data, including distance error by audio

condition and spatial test score distributions. Apart from the visibly lower distance error in the spatial audio condition shown only by females, a corresponding difference in effect size (Cohen's d) was observed between the genders; females exhibited a medium effect ($d = 0.53$) whilst males exhibited negligible effect ($d = 0.036$). Considering the higher proportion of low-spatial ability participants in the female distribution (Figure 1, bottom), well-studied sex and gender differences in spatial ability [13], these results are in line with existing literature demonstrating more positive effect of cognitive support on low spatial ability individuals than on those with high spatial ability [4].

DISCUSSION AND FUTURE WORK

The results of the study indicate that spatial familiarization in VR with spatial audio cues results in higher spatial recall accuracy compared to a lack thereof. Participants in the spatial audio condition demonstrated statistically significant lower recall error when mapping chair locations, and we know this is based on successful intuitive usage of the audio cue as they were not told of the presence of spatial audio. Based on this result, its incorporation into VR interfaces can be a cost-effective and intuitive addition to support more accurate spatial knowledge acquisition in VR. The implications for future virtual reality design given by this include understanding that integrating spatial audio into VEs not just in a general sense, but specifically in an element-bound, area-specific form, helps users to develop a more refined spatial understanding of the environment.

Additionally, the trending results on gender differences and spatial ability relation to responsiveness to spatial audio cues afford some insight as to how intelligent VR interfaces that originally do not need sound as part of default application may support dynamic inclusion of spatial audio cues in response to the state of a user's performance on spatially-oriented activities.

Future work can involve comparing the performance of spatial audio cue-augmented cognitive mapping with that of mapping a real-world environment to determine the practical sustainability of such a mapping. In such a case, we also wish to evaluate and compare measured performance and subject satisfaction in relation to actual activities associated with the spatial mapping generated from these environments (e.g. schematic map generation).

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